

LOW ACRYLONITRILE NBR PATENT LANDSCAPE REPORT

1. ABSTRACT

This report evaluates the provided patent evidence against the target of low-acrylonitrile nitrile butadiene rubber (NBR). The patent manifest contains one primary patent, US10465031B2, which is not a synthesis patent for NBR but a downstream elastomer article patent focused on fluorinated nitrile rubber-based packers and seals for sour-oil and blowout-preventer service. The evidence explicitly discusses NBR, HNBR, and XNBR as elastomer families and notes that conventional NBR formulations vary acrylonitrile (ACN) and butadiene ratios to tune oil/fuel resistance and low-temperature properties, but it does not disclose a preparation route for low-ACN NBR itself.

Because the patent is application-focused rather than polymerization-focused, the landscape contribution is indirect: it confirms the industrial relevance of ACN content as a key composition variable in nitrile elastomers and provides comparative performance data for 17% ACN HNBR and 36% ACN HNBR in H₂S permeability simulations. The disclosed examples center on fluorocarbon grafting of HNBR latex using Capstone MS-62 and tert-butyl hydroperoxide (TBHP) initiator, followed by drying and thermal grafting. These examples demonstrate that fluorination/grafting can reduce gas solubility, diffusion, and permeability, but they do not address low-ACN NBR synthesis.

Overall, the evidence supports the broader technical context that ACN level is a critical design parameter in nitrile elastomers, yet it does not provide direct low-acrylonitrile NBR copolymerization conditions, monomer feed ratios, or emulsion polymerization recipes. For low-ACN NBR synthesis, the most relevant actionable parameters remain those absent here: controlled butadiene/acrylonitrile feed ratio, semi-continuous or starved-feed emulsion polymerization, and explicit targeting of low bound acrylonitrile content.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US10465031B2
- Patent Title: Short-chain fluorocarbon-grafted elastomer blowout preventer packers and seals
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- Assignee: Not disclosed
- Jurisdiction: US

- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent does not disclose synthesis of low-acrylonitrile NBR. It describes fluorocarbon grafting onto nitrile rubber-based elastomers, especially HNBR latex, by adding Capstone MS-62 to latex and carrying out thermal grafting after drying, with TBHP used as initiator. - Compound composition: between 50 and 60% nitrile rubber — Claims 1 and 7 - Carbon black content: between 25 to 35% — Claims 1 and 7 - Fluorine-to-carbon ratio: about 0.10 to 0.18 — Claim 5 - Alternative fluorine-to-carbon ratio: about 0.18 to 0.36 — Source text - HNBR latex polymer content: approximately 40 wt % — Example II - Initiator: tert-butyl hydroperoxide (TBHP) 70 wt % solution in water — Example II - H₂S sorption pressure range: 5 kPa to 100 kPa — Example I - GCMC equilibration steps: 100,000 steps — Example I - GCMC production run: 1,000,000 steps — Example I - Capstone loading levels: 0.00%, 2.00%, 4.00%, 6.00% — Tables 3 and 4

Relevant Experimental Evidence - Example I: H₂S permeability was compared for 17% ACN HNBR and 36% ACN HNBR, showing that higher ACN content reduced diffusion and permeability. - Example I: Grafting Capstone chains further decreased solubility, diffusion coefficient, and permeability. - Example II: Capstone MS-62 was added to HNBR latex and thermally grafted after drying, using TBHP as initiator. - Example II: The example used Zeon Chemical HNBR latex containing about 40 wt % polymer. - Example II: The disclosure links fluorocarbon grafting to improved H₂S resistance in nitrile elastomers.

Technical Relevance This patent is only indirectly relevant to low-acrylonitrile NBR because it confirms that ACN content strongly affects nitrile elastomer transport properties and performance in sour environments. However, it does not disclose NBR polymerization or low-ACN composition control, so it is not a direct synthesis reference.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

The provided evidence does not disclose a direct low-acrylonitrile NBR synthesis patent. Instead, US10465031B2 is an application patent for fluorinated nitrile elastomer packers and seals, with indirect relevance through its discussion of ACN-dependent properties in nitrile elastomers. For

low-ACN NBR synthesis, the evidence suggests the key missing process variables are controlled acrylonitrile/butadiene feed ratio, emulsion or semi-continuous copolymerization, and explicit targeting of bound acrylonitrile content below typical commercial levels. No validated synthesis recipe for low-ACN NBR can be extracted from this patent alone.

5. REFERENCES

- US10465031B2 | Short-chain fluorocarbon-grafted elastomer blowout preventer packers and seals ... | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10465031B2/en>